

MATH 195: 10/8
CHANGE OF BASE WITH LOGARITHMS AND EXPONENTIAL FUNCTIONS

An important fact is that all exponential functions and all logarithms are essentially the same, so that for many purposes it doesn't matter which base you use.

Change of base formulas.

Consider positive bases a and b , neither of which is 1. These formulas tell you how to convert from base a to base b .

- $a^x = b^{\log_b(a)x}$
- $\log_a(x) = \frac{\log_b(x)}{\log_b(a)}$

These often appear in converting to base e .

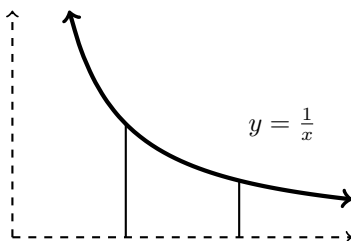
- $a^x = e^{\ln(a)x}$
- $\log_a(x) = \frac{\ln(x)}{\ln(a)}$

Many calculators will have buttons for the natural logarithm and common logarithm, but not other bases. The change of base formula for logarithm can be used to calculate logarithms of any base.

In calculus it is common to use only base e , as that is the base which is nicest when doing calculus. Here is an example to help illustrate why base e is the natural base for logarithms.

The quadrature of the hyperbola.

In 1649, the mathematician Alphonse de Sarasa used a clever geometric argument to determine how to find a rectangle with the same area as under the hyperbola $y = \frac{1}{x}$. In modern terminology, we call this the *integral* of $\frac{1}{x}$.



This work uncovered a remarkable fact. Let $A(u)$ denote the area under $y = \frac{1}{x}$ from $x = 1$ to $x = u$. Then,

$$A(uv) = A(u) + A(v).$$

This should look a lot like one of the rules for logarithms

$$\log_a(uv) = \log_a(u) + \log_a(v).$$

Indeed, circa 80 years later mathematician Leonhard Euler noticed a number e with the property that $A(e) = 1$. And it turns out that $A(u) = \ln(u)$ is exactly the natural logarithm.

The lesson is, if you care about integrals—one of the most important concepts in calculus—then base e is a natural choice of base for the logarithm.

The change of base formulas can be used to solve equations with exponentials of different bases on each side.

$$a^A = b^B$$

$$\log_a A = \log_b B$$

Namely:

- Rewrite the equation so each side has the same base.
- Then, hit both sides with the same inverse function to cancel out the exponentials (or logarithms).

Alternatively for exponential equations you can organize the work in the opposite order.

- Hit both sides with the same logarithm.
- Use rules for logarithms to rewrite as an equivalent equation without any exponents.

Often other rules for logarithms and exponentials are useful.

PRACTICE PROBLEMS

For all problems, please simplify as much as possible.

- (1) Solve the equation $2^x = 2^{3-2x}$. Give an exact answer not in terms of logarithms.
- (2) Solve the equation $2^{x-1} = 4^{2+x}$. Give an exact answer not in terms of logarithms.
- (3) Solve the equation $2^{3x+1} = 3^{4-x}$. Give an exact answer, but it will have to be in terms of logarithms.
- (4) Find all points where the graphs of $f(x) = e^{-x}$ and $g(x) = 4^{x-4}$ intersect. Give both the x - and y -coordinates.
- (5) Solve the equation $\log_2(x) = \log_2(2x - 4)$.
- (6) Solve the equation $\log_2(x) = 2 - \log_2(x)$. *[Hint: hit both sides with the exponential function with base 2, then use rules for exponents to simplify.]*
- (7) Solve the equation $\log_2(x) = \log_3(4 - x)$.
- (8) Explain why the equation $\log_2(x) = \log_2(-x)$. Can you give a geometric explanation?
- (9) Solve the equation $\ln(x^2) = \ln(4x + 12)$.
- (10) Solve the equation $\ln(x^2 - 12) = \ln(4x)$. *[Hint: although this problem reduces to solving the same quadratic equation as the previous problem, it has one fewer solution. Remember what you know about domains of logarithms!]*
- (11) Give a general explanation of what you should do when solving equations with logarithms to confirm a “solution” really is in the domain of the equation.